

Unit 3(Part 1)

Advanced Networking and Routing Protocols: Advanced IP addressing, routing protocols including RIP, OSPF, and BGP, multicast routing and protocols.

Routing Protocols

Routing protocols help routers exchange information and choose the best path for data transmission.

Types of Routing Protocols:

- **Interior Gateway Protocols (IGP)** – used within a single network (Autonomous System)
 - RIP
 - OSPF
- **Exterior Gateway Protocols (EGP)** – used between different networks
 - BGP

1. RIP (Routing Information Protocol)

- RIP is a **Distance Vector Routing Protocol**.
- It uses hop count as a metric to find the best path.

Working

- Each router sends its entire routing table every 30 seconds.
- Path with minimum hops is selected.
- Maximum hop count = 15 (16: unreachable).
- Broadcasts (RIPv1) or multicasts (RIPv2/RIPng) routing information.

Algorithm: Uses Bellman-Ford Algorithm

Advantages

- Simple and easy to configure
- Low resource usage
- Works on almost all devices
- Suitable for small networks

Disadvantages

- Limited scalability (15-hop limit)
- Slow convergence
- Sends full routing table → high bandwidth usage
- No advanced features like hierarchy or efficient load balancing

Types of RIP Versions

Feature	RIPv1 (1988)	RIPv2 (1993)	RIPng (1997) for IPv6
Updates sent via	Broadcast (255.255.255.255)	Multicast (224.0.0.9)	Multicast (FF02::9)
Addressing	Classful (no subnet info)	Classless (includes subnet mask)	Classless (IPv6 only)
Authentication	Not supported	Supported	Supported
IP version supported	IPv4 only	IPv4 only	IPv6 only

Where is RIP Used?

- Small to medium-sized networks with simple routing needs.
- Legacy networks that were built before OSPF/EIGRP became standard.
- Educational labs for training and learning routing basics.
- Backup routing protocol in case the primary protocol fails.

2. OSPF (Open Shortest Path First)

- OSPF is a **Link-State Routing Protocol**.
- Uses network topology information to determine the best route.

Working

- Routers build a topology map (link-state database).
- Only changes/updates are sent (not full table).
- Uses cost (bandwidth) as metric.

Algorithm

- Uses Dijkstra's Shortest Path Algorithm

Advantages

- Fast convergence
- Highly scalable (no hop limit)
- Supports hierarchical design (areas)
- Efficient bandwidth usage
- Supports load balancing (ECMP)

Disadvantages

- Complex to configure
- Requires more CPU & memory
- Higher initial setup effort

Key Differences Between RIP and OSPF

Feature	RIP	OSPF
Full Form	Routing Information Protocol	Open Shortest Path First
Type	Distance Vector	Link-State
Algorithm	Bellman-Ford	Dijkstra
Metric	Hop Count	Cost (Bandwidth)
Max Hops	15	No limit
Network Size	Small networks	Large networks
Convergence	Slow	Fast
Updates	Full routing table	Only changes
Bandwidth Usage	High	Low
Complexity	Simple	Complex
Administrative Distance	120	110
Protocol Used	UDP	IP
Scalability	Poor	High
Hierarchical Design	Not supported	Supported

3. Border Gateway Protocol (BGP)

- BGP (Border Gateway Protocol) is a **routing protocol** used to exchange **routing information** between different networks (**Autonomous Systems – AS**)
- It is an Exterior Gateway Protocol (EGP)
- It is a Path Vector Protocol

BGP decides how data travels across the internet between large networks (like ISPs).

Autonomous System (AS)

- A group of networks under one organization (like Airtel, Jio, etc.)

Path Vector Routing

- BGP stores the path (list of AS) to reach a destination
- Helps avoid loops and choose the best route

Main Functions of BGP

BGP peers perform 3 functions, which are given below:

1. Establish connection (via TCP)

- The first function consists of initial peer acquisition and authentication. Both the peers established a TCP connection and perform message exchange that guarantees both sides have agreed to communicate.

2. Exchange routing information

The second function mainly focuses on sending negative or positive reachability information.

3. Maintain connection (check if alive)

The third function verifies that the peers and the network connection between them are functioning correctly.

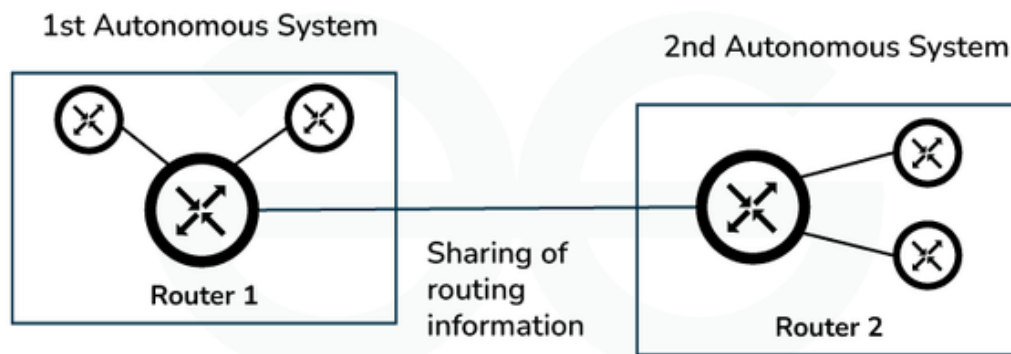
Features of BGP

- Works between different AS (inter-domain routing)
- Uses TCP protocol (reliable communication)
- Supports policy-based routing (admin controls routes)
- Provides high scalability (handles internet-sized networks)
- Supports CIDR & route aggregation
- Conserves bandwidth and improves efficiency

Types of BGP

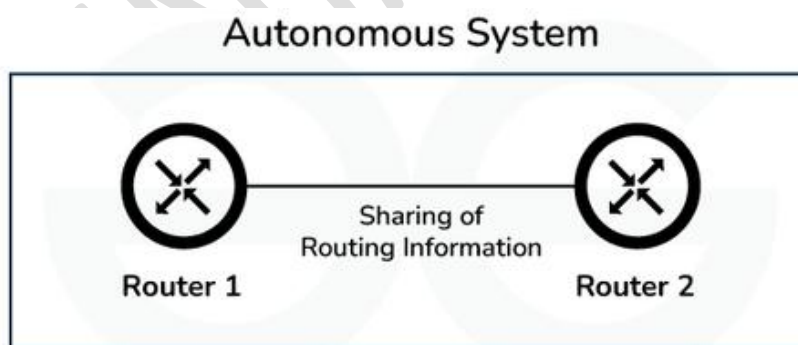
1. External BGP (eBGP)

- It is used to interchange routing information between the routers in different autonomous systems, it is also known as eBGP(External Border Gateway Protocol).
- Used between different AS
- Example: ISP ↔ ISP



2. Internal BGP (iBGP)

- It is used to interchange routing information between the routers in the same autonomous system, it is also known as iBGP(Internal Border Gateway Protocol). Internal routers also ensure consistency among routers for sharing routing information
- Used within the same AS
- Ensures all routers share same routing info



Why BGP is Important

- Backbone of the Internet
- Handles a huge number of routes (scalable)
- Supports multihoming (multiple ISPs)

- Selects the best path for data transfer
- Provides basic control and reliability

Difference Between BGP and OSPF

BGP	OSPF
It follows the Path Vector Routing Algorithm	It follows the Link State Routing Algorithm
The speed of convergence is very slow in BGP	The speed of convergence is fast in the case of OSPF
BGP is also called inter-domain routing protocol	OSPF is also called intra-domain routing protocol
In BGP routing operation is performed between two AS	In OSPF routing operation is performed inside an AS
In BGP, TCP protocol is used	In OSPF, IP protocol is used

RIP vs OSPF vs BGP

Feature	RIP	OSPF	BGP
Type	Distance Vector	Link-State	Path Vector
Use	Small networks	Enterprise networks	Internet (ISP level)
Metric	Hop count	Cost (bandwidth)	Path attributes
Speed	Slow	Fast	Medium/Slow
Max size	Small	Large	Very large (global)
Transport	UDP	IP (protocol 89)	TCP (port 179)

Updates	Periodic	Triggered	Incremental
Complexity	Easy	Medium	Hard

MULTICAST ROUTING

- Multicast routing is a method where a **single sender transmits data to multiple selected receivers (a group) using the most efficient path possible.**
- Instead of sending separate copies (like unicast), the network:
 - Sends one copy
 - Duplicates it only where needed

Why Multicast Routing is Needed?

Problems without Multicast

- Sender sends multiple copies in unicast
- High bandwidth consumption
- Network congestion increases
- Not scalable for large systems

Advantages

- Efficient bandwidth usage
- Reduces redundant transmissions
- Suitable for real-time applications
- Improves overall network performance

The Key Component of IP Multicast

1. Multicast Group

A multicast group is a **dynamic set of devices that are interested in receiving the same data stream and a unique multicast IP address defines each group**

so the network Node interested in receiving data forwarded to that multicast IP address needs to become a member of that multicast group. devices join or leave Multicast Group dynamically.

2. Source

Source is the device that **originate the data packets which is sent to Multicast Group , any device that has the capacity of generating multicast traffic can be a source.** there can be more then one source which are sending data to a multicast group and this is useful in case where multiple source need data to be distributed among same set of recipients.

3. Internet Group Management Protocol (IGMP)

Internet Group Management Protocol (IGMP) is used to **manage host group membership in IPv4 networks allow the host to report their multicast group memberships to required routers.**

Multicast Listener Discovery (MLD) is same as Internet Group Management Protocol (IGMP) just it is used for IPv6 networks and manage membership for IPv6 host

4. Multicast Router

Multicast Router is a **networking device responsible for managing and directing multicast traffic efficiently from the source to destination**, they use Internet Group Management Protocol (IGMP) to locate receiver of the active group , Multicast Router also are responsible for building and maintaining distribution trees and mapping the source to the respected receivers on the network.

5. Receiver

Receiver is an end point device that has joined the multicast group and has subscribed to receive the data packets, to join the multicast group receiver need to configure itself mainly using using Internet Group Management Protocol (IGMP) so it **only receives data packets which are send to multicast group in which it belongs to.**

How Does IP Multicast Works?

1. First of all device interested in receiving data **join the multicast group which is identified by unique multicast IP address.**
2. Now once the group is joined by receiver the source device craft the data packets and forward them to multicast group address not to any individual receiver.
3. Now the **Multicast router** which is aware of the Multicast group using IGMP efficiently forward multicast packets to the the branch of network where the interested users are located and here in order to optimize the delivery path router use distribution trees to map the source to receivers.
4. Once the data packets reach the branch of network **the Receiver which have joined the multicast group receive and process the data packets** while other devices in network branch ignore the multicast traffic

Benefits of Multicast

- **Efficiency:** Drastically reduces network traffic and CPU load by sending a single stream instead of multiple unicast streams.
- **Scalability:** Ideal for applications with large audiences like live streaming, reducing the strain on source servers.
- **Improved Experience:** Lower latency due to reduced congestion.

Class D Range (IPv4)

Class D is part of the IPv4 address space, and the range reserved for multicast addresses is:

- **224.0.0.0 to 239.255.255.255**

Any IP address within this range is used for multicast communication. Devices or hosts wishing to communicate with a multicast group must join the specific address in this range. This range is distinct from other address classes like:

- **Class A (1.0.0.0 - 127.255.255.255):** [Unicast addresses]

- **Class B (128.0.0.0 - 191.255.255.255):** [Unicast addresses]
- **Class C (192.0.0.0 - 223.255.255.255):** [Unicast addresses]
- **Class E (240.0.0.0 - 255.255.255.255):** [Reserved for future use and experimental purposes]

Special Multicast Addresses within Class D:

- **224.0.0.0 to 224.0.0.255:** These addresses are used for **link-local multicast**. They are reserved for multicast communication on the local network (no routing across routers).

For example, **224.0.0.1** is the multicast address for all hosts on a local network.

- **233.0.0.0 to 233.255.255.255:** These addresses are used for **source-specific multicast (SSM)**, where the multicast group is associated with a particular source.
- **239.0.0.0 to 239.255.255.255:** This range is designated for administratively scoped multicast, which is used for **private, non-routable multicast groups**.

Multicast Tree Concept

Structure

- Root node is sender
- Intermediate nodes are routers
- Leaf nodes are receivers

Properties

- Loop-free structure
- Ensures each receiver gets one copy
- Optimized path reduces delay

Types of Multicast Trees

1. Source-Based Tree (Shortest Path Tree)

- Separate tree for each source
- Built using shortest path algorithm

Working

- Source starts sending data
- Routers calculate shortest path using routing tables
- Tree grows outward from source

Advantages

- Minimum delay
- Efficient routing path
- No unnecessary intermediate nodes

Disadvantages

- High control overhead
- Not scalable with many senders
- Requires more memory in routers

Example

- MOSPF

2. Shared Tree

- Single tree shared among all senders
- Root node is Rendezvous Point(RP)

Working

- Receivers send join request to RP
- RP maintains group membership
- Senders send data to RP

- RP distributes data to all members

Advantages

- Reduced overhead
- Easier to manage
- Scalable for large networks

Disadvantages

- Longer path possible
- RP failure affects communication

Example

- PIM-SM

Multicast Routing Techniques

These are methods used by routers to **efficiently deliver multicast data** from sender to multiple receivers.

1. Flooding

- Packet is sent to **all routers in the network**
- Every router forwards packet to all its neighbors

Working

1. Source sends multicast packet
2. Packet reaches all neighboring routers
3. Each router forwards to all neighbors except incoming link
4. Process continues until all routers receive packet

Problem

- Duplicate packets
- High bandwidth usage
- Loop formation possible

Example

- Network has routers A, B, C, D
- A sends packet
- B, C, D all receive
- Even routers with no receivers still get data
- **Used as initial step in Dense Mode**

2. Reverse Path Forwarding (RPF)

- Router forwards packet only if it comes from **shortest path to source**

Working

1. Router receives packet
2. Checks routing table for shortest path to source
3. Compares incoming interface
 - If correct → forward
 - If wrong → discard

Purpose

- Prevent loops
- Avoid duplicate packets

Example

- Router R has shortest path to source via interface X
- Packet arrives from interface Y
- Router discards packet

Importance

- Core technique in almost all multicast protocols

3. Pruning

- Stops multicast traffic in branches where **no receivers exist**

Working

1. Router receives multicast data
2. Checks for group members
3. If none:
 - Sends PRUNE message to upstream router
4. Upstream router stops forwarding

Purpose

- Reduce unnecessary traffic
- Improve efficiency
- Used in **Sparse Mode**

Example

- Router B has no receivers
- Sends prune message to A
- A stops sending data to B

4. Grafting

- Reconnects a branch that was pruned earlier

Working

1. New receiver joins
2. Router sends GRAFT message upstream

3. Upstream router resumes forwarding

Purpose

- Allow dynamic joining
- Restore multicast path

Example

- Branch was pruned earlier
- New user joins
- Router sends graft
- Data starts flowing again

5. Join Mechanism

- Receivers explicitly request multicast data

Working

1. Receiver wants to join group
2. Sends JOIN message
3. Routers build path to source or RP
4. Receiver becomes part of multicast tree

Purpose

- Ensure only interested nodes receive data
- Used in **Sparse Mode**

Example

- Student joins online class
- Only then receives lecture

6. Rendezvous Point (RP)

- Central router in Sparse Mode
- Meeting point for sender and receivers

Working

1. Receivers send JOIN to RP
2. Sender sends data to RP
3. RP forwards data to receivers

Purpose

- Simplifies multicast tree creation
- Centralizes communication

Example

- RP acts like server
- All users connect to it
- Data distributed from there

Problem: Can become bottleneck, Single point of failure

7. Shortest Path Switching (SPT Switch)

- Switch from shared path to direct shortest path

Working

1. Initially data flows via RP
2. Router detects delay
3. Builds direct path to source
4. Stops using RP path

Purpose

- Reduce delay
- Improve performance

Example

- Initially: Sender → RP → Receiver
- Later: Sender → Receiver directly
- Used in Sparse Mode (PIM-SM)

Dense Mode Techniques

- Flooding
- Reverse Path Forwarding
- Pruning
- Grafting

Sparse Mode Techniques

- Join mechanism
- Rendezvous Point
- Shortest Path Switching

Types of Multicast Routing Modes

1. Dense Mode

- Dense Mode is a multicast routing approach
- Assumes **most routers in the network have receivers**
- Data is sent to all routers initially
- “Send data everywhere first, then remove unwanted parts”
- Also called **Push Model**

Key Characteristics

- Uses flooding to distribute data
- Builds source-based tree (Shortest Path Tree)
- Does not require prior knowledge of receivers
- Uses Reverse Path Forwarding (RPF)
- Uses pruning and grafting

Working of Dense Mode

Step 1: Initial Flooding

- Source sends multicast packet
- Packet is forwarded to all routers in network
- Every router receives the packet

Step 2: Reverse Path Forwarding (RPF)

- Each router checks:
 - Did packet come from shortest path to source?
 - If YES: Forward packet
 - If NO: Discard packet

Step 3: Duplicate Control

- RPF ensures: No loops
- No duplicate packets

Step 4: Pruning

- Routers check if they have receivers
 - If no receivers: Send PRUNE message upstream
- Upstream router stops sending data

Step 5: Multicast Tree Formation

- After pruning:
 - Only required branches remain
- Final tree connects source to receivers

Step 6: Grafting (Dynamic Join)

- If new receiver joins:
 - Router sends GRAFT message
- Branch is reactivated
- Data flow resumes

Example

- Source S
- Routers A, B, C, D, E
- Only C and E have receivers

1. S sends packets
2. A, B, C, D, E all receive (flooding)
3. A, B, D have no receivers → send prune
4. Only C and E continue receiving
5. Tree forms: $S \rightarrow C$ and $S \rightarrow E$

If New Receiver Joins

- Suppose B gets a receiver
- B sends graft
- Data starts flowing again to B

Advantages of Dense Mode

- **Simplicity:** Easy to implement
- **Fast Initial Delivery:** No delay in setup
- **No Membership Requirement:** No need to know receivers beforehand

Disadvantages of Dense Mode

- **High Bandwidth Usage:** Flooding sends unnecessary data
- **Not Scalable:** Poor performance in large networks
- **Inefficient for Sparse Groups:** Wasteful when few receivers exist

Protocols Using Dense Mode

- **DVMRP:** Distance Vector based multicast protocol
- **PIM-DM:** Protocol Independent Multicast Dense Mode

2. Sparse Mode

- Sparse Mode is a multicast routing approach
- Assumes only few routers have receivers
- Data is sent only when receivers request it

- “Send data only to those who ask for it”
- Also called Pull Model

Key Characteristics

- Uses join-based mechanism
- Uses a central router called Rendezvous Point (RP)
- Initially builds shared tree
- May switch to source-based tree (Shortest Path Tree)
- Efficient use of bandwidth
- Suitable for large networks

Working of Sparse Mode

Step 1: Receiver Joins Group

- A receiver wants multicast data
- Sends **JOIN message** towards RP
- Path is created from receiver to RP

Step 2: Tree Formation (Shared Tree)

- As JOIN messages propagate
- Routers form a shared multicast tree
- Root of tree is RP

Step 3: Sender Sends Data

- Sender sends multicast data to RP
- Data travels using unicast routing

Step 4: RP Distributes Data

- RP forwards data to all receivers
- Data flows along the shared tree

Step 5: Shortest Path Switching (SPT)

- Routers may switch to direct path from source

- Creates Shortest Path Tree
- Reduces delay and improves performance

Step 6: Continuous Data Delivery

- Data continues flowing only to joined receivers
- No unnecessary transmission

Techniques Used in Sparse Mode

- Join Mechanism
- Rendezvous Point (RP)
- Shared Tree
- Shortest Path Switching

Example

- Source S
 - Routers A, B, C, D, E
 - Only C and E have receivers
 - RP is router A
1. C and E send JOIN to RP (A)
 2. Path forms: $C \rightarrow A$ and $E \rightarrow A$
 3. Sender S sends data to A
 4. A forwards data to C and E
 - Later: C and E may switch to direct path from S

Advantages of Sparse Mode

- **Efficient Bandwidth Usage:** Data sent only where needed
- **Scalable:** Works well for large networks
- **Suitable for Sparse Groups:** Best when few receivers exist

Disadvantages of Sparse Mode

- **Initial Delay:** Time needed to join group
- **RP Dependency:** RP failure affects communication

- **Complexity:** More complex than dense mode

Protocol Using Sparse Mode

PIM-SM: Most widely used multicast protocol

MULTICAST ROUTING PROTOCOLS

1. DVMRP: Distance Vector Multicast Routing Protocol

- DVMRP is a multicast routing protocol used to send data to multiple receivers
- It is based on **distance vector routing algorithm**
- Works mainly in **dense mode multicast routing**
- Assumes most routers in the network want multicast data
- Uses flooding + pruning approach
- Builds multicast trees dynamically

Important Characteristics

- Uses **Reverse Path Forwarding (RPF)**
- Creates **source-based trees (Shortest Path Trees)**
- Maintains its own routing table
- Works independently of unicast routing protocols
- Uses hop count as metric
- Periodically exchanges routing updates

Components of DVMRP

Routing Table

- Stores shortest path to all destinations

- Based on distance vector algorithm

Multicast Forwarding Table

- Keeps track of multicast group membership
- Maintains information about active branches

Working of DVMRP

Step 1: Initial Flooding

- Source sends multicast packets
- Packets are forwarded to all routers in network

Step 2: Reverse Path Forwarding Check

- Each router checks:
 - Whether packet came from shortest path to source
 - If YES: Forward packet
 - If NO: Discard packet

Step 3: Duplicate Packet Control

- RPF ensures:
 - No loops
 - No duplicate forwarding

Step 4: Pruning Process

- Routers without receivers send **PRUNE** message
- Stops packet forwarding in that branch

Step 5: Formation of Multicast Tree

- Only branches with receivers remain active
- Final tree is a Shortest Path Tree from source

Step 6: Grafting (When Needed)

- If a new receiver joins:

- Router sends **GRAFT message**
- Branch is reactivated

Step 7: Periodic Updates

- Routers exchange routing information periodically
- Ensures updated network topology

Example

- Network has 8 routers
- Only 3 routers have receivers

Flow:

1. Source floods packets to all 8 routers
2. 5 routers send prune messages
3. Only 3 routers continue receiving
4. Tree forms connecting source to those 3 routers

Advantages of DVMRP

- Simple and easy to implement
- Automatically builds multicast trees
- Works well in small and dense networks
- No need for prior group membership knowledge

Disadvantages of DVMRP

- High bandwidth usage due to flooding
- Not scalable for large networks
- Maintains separate routing table (extra overhead)
- Periodic updates increase traffic
- Inefficient when few receivers exist

Where DVMRP is Used

- Early multicast networks

- MBone (Multicast Backbone)
- Small or experimental networks

2. PIM (Protocol Independent Multicast)

- PIM is a multicast routing protocol used to deliver data to multiple receivers
- It is called **protocol independent** because it does not depend on any specific unicast routing protocol
- It uses existing routing tables (like OSPF, RIP, BGP)
- PIM separates **multicast routing** from **unicast routing**
- It uses unicast routing information to build multicast trees
- **Supports both Dense Mode and Sparse Mode**

Types of PIM

- PIM-DM (Dense Mode)
- PIM-SM (Sparse Mode)

1. PIM Dense Mode (PIM-DM)

- Assumes **most routers have receivers**
- Uses **flooding approach**

Working

1. Source sends multicast packets
2. Packets are flooded to all routers
3. Routers apply Reverse Path Forwarding check
4. Routers without receivers send prune messages
5. Unnecessary branches are removed
6. Multicast tree is formed

Mechanisms Used

- **Reverse Path Forwarding**
- **Pruning**
- **Grafting**

Characteristics

- Builds source-based tree
- Uses periodic flooding
- No central control point

Advantages

- Simple to implement
- Quick initial setup
- No need to maintain group membership initially

Disadvantages

- High bandwidth consumption
- Inefficient for large networks
- Not suitable when receivers are few

2. PIM Sparse Mode (PIM-SM)

- Assumes **few routers have receivers**
- Uses **join-based approach**
- Data sent only to interested receivers

Key Component: Rendezvous Point (RP)

Working

1. Receiver sends join message towards RP
2. Path is established from receiver to RP
3. Sender sends multicast data to RP
4. RP forwards data to all joined receivers
5. Routers may switch to shortest path

Shortest Path Switching

- After initial communication
- Routers create direct path from sender
- Reduces delay and improves efficiency

Characteristics

- Uses **shared tree initially**
- May switch to **source-based tree**
- Efficient bandwidth usage

Advantages

- Scalable for large networks
- Efficient for sparse distribution
- Reduces unnecessary traffic

Disadvantages

- Initial delay due to RP
- RP can become bottleneck
- More complex than PIM-DM

COMPARISON: DVMRP vs PIM

DVMRP	PIM
Distance Vector Multicast Routing Protocol	Protocol Independent Multicast
Works only in Dense Mode	Works in both Dense Mode and Sparse Mode
Uses flood and prune approach	Uses flood (DM) and join-based approach (SM)
Maintains its own routing table	Uses existing unicast routing table
Based on distance vector routing	Independent of routing protocol
Builds only source-based tree	Builds source-based tree (DM) and shared + shortest path tree (SM)
Uses prune and graft messages	Uses join, prune, and register messages
High bandwidth consumption due to flooding	Efficient bandwidth usage (especially in Sparse Mode)
Not scalable for large networks	Highly scalable for large networks
Simple to implement	More complex than DVMRP
Does not use Rendezvous Point	Uses Rendezvous Point in Sparse Mode
Mainly used in early multicast networks	Widely used in modern networks
Less flexible	More flexible

COMPARISON: DENSE MODE vs SPARSE MODE

Dense Mode	Sparse Mode
Assumes most routers have receivers	Assumes very few routers have receivers
Uses push model (send data first)	Uses pull model (send on request)
Uses flooding to send packets to all routers	Uses join mechanism to build multicast tree
Pruning is used to remove unwanted branches	No need for pruning initially
Builds source-based tree (Shortest Path Tree)	Builds shared tree first, then may switch to shortest path tree
No central control router required	Uses Rendezvous Point (RP)
High bandwidth usage due to flooding	Efficient bandwidth usage
Fast initial data delivery	Initial delay due to join process
Less scalable for large networks	Highly scalable for large networks
Suitable for small or dense networks	Suitable for large and sparse networks
Uses techniques like flooding, RPF, pruning, grafting	Uses join, RP-based forwarding, shortest path switching
Example: PIM-DM, DVMRP	Example: PIM-SM

COMPARISON: PIMDM vs PIMSM

PIM-DM (Dense Mode)	PIM-SM (Sparse Mode)
Protocol Independent Multicast - Dense Mode	Protocol Independent Multicast - Sparse Mode
Assumes most routers have receivers	Assumes few routers have receivers
Uses push model (send first, then prune)	Uses pull model (join first, then send)
Uses flooding to send packets to all routers	Uses join messages to build multicast tree
Pruning removes unnecessary branches	No initial flooding, so minimal pruning
Builds source-based tree (Shortest Path Tree)	Builds shared tree first, then may switch to shortest path tree
No Rendezvous Point used	Uses Rendezvous Point (RP)
High bandwidth usage due to flooding	Efficient bandwidth usage
Fast initial data delivery	Slight delay due to join process
Less scalable	Highly scalable
Suitable for small/dense networks	Suitable for large/sparse networks
Uses RPF, pruning, grafting	Uses join, RP, register, shortest path switching
Simpler to implement	More complex than PIM-DM